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Student Mental Health Analysis: A Classification Approach for Predicting Depression Levels

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Classification Analysis Report

Abstract

The goal of this paper is to forecast depression levels among students using categorization algorithms. This analysis was carried out using the Student Depression Dataset, which contains markers of financial hardship and academic accomplishment that may be linked to depression levels. The process is as follows: exploratory data analysis (EDA), model construction using logistic regression, hyper-parameter optimization, and feature selection. The Logistic Regression model demonstrated promising accuracy on both the training and test datasets, highlighting the importance of feature engineering and data cleaning in enhancing classification performance.

1. Introduction

1.1 Problem Statement

The purpose of this project is to predict depression levels in students using a range of factors such as financial stress, academic performance, and demographic information. Accurate prediction of depression levels can benefit educational institutions and mental health practitioners in providing early intervention and care.

1.2 Dataset

This analysis makes use of the Student Depression Dataset, which was made public through Kaggle. It focuses on mental health and well-being (Goal 3), which aligns with the United Nations Sustainable Development Goals.

Dataset Details:

- **Number of Samples:** 27,901 entries
- **Number of Features:** 18 attributes, including demographic, academic, and stress-related factors
- **Target Variable:** Depression level (categorized as "low," "medium," and "high")
- **Missing Data:** Financial Stress had 15% missing values, imputed using the median value.
- **Source:** Kaggle (uploaded two months ago by Shodolamu Opeyemi) [1]

1.3 Objective

The major goal is to create a predictive classification model that evaluates the likelihood of depression among students based on specific parameters.

2. Methodology

2.1 Data Preprocessing

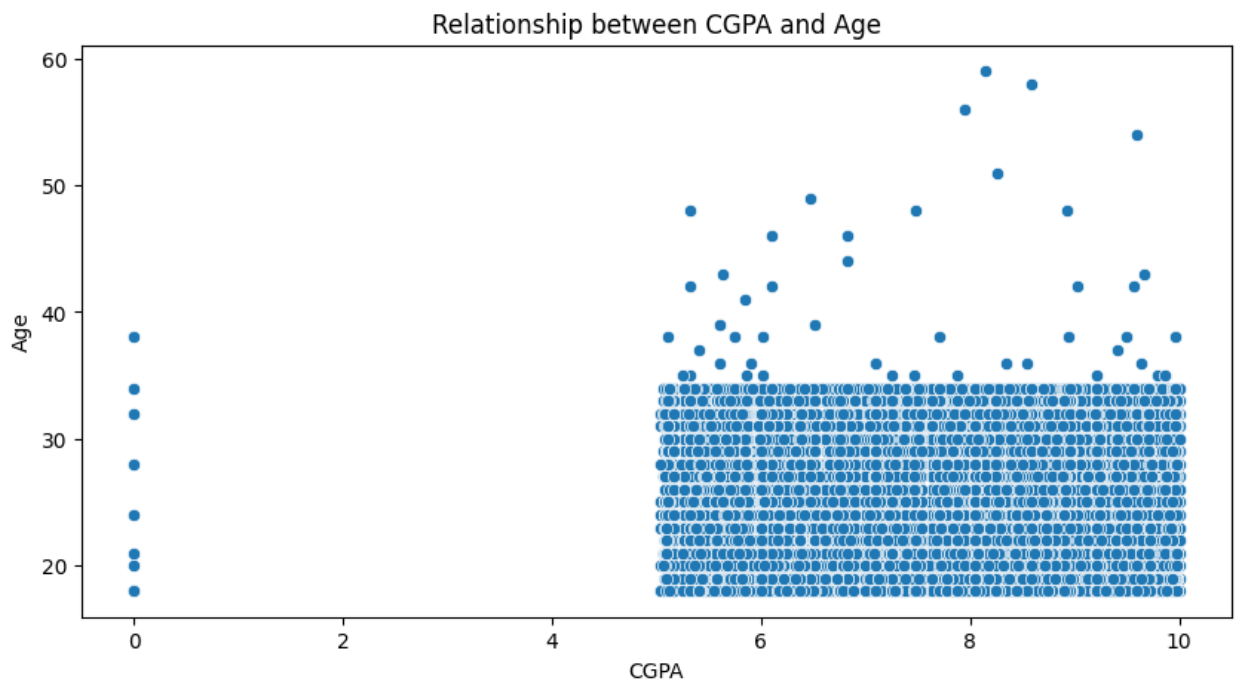
To ensure accurate and efficient model building, the following preprocessing steps were undertaken:

- **Handling Missing Values:** Imputed missing values in the "Financial Stress" feature using the median.
- **Data Type Conversion:** Converted features such as "Age" and "CGPA" to appropriate data types (integer and float, respectively).
- **Duplicate Removal:** Eliminated duplicate entries.
- **Outlier Treatment:** While not explicitly conducted, future iterations should address potential outliers in stress and academic features.
- **Data Normalization:** Recommended to enhance classification model performance.

2.2 Exploratory Data Analysis (EDA)

EDA was conducted using summary statistics and visualizations to understand the dataset.

- **Scatter Plot:** Explored the relationship between CGPA and Age, revealing potential trends [2].



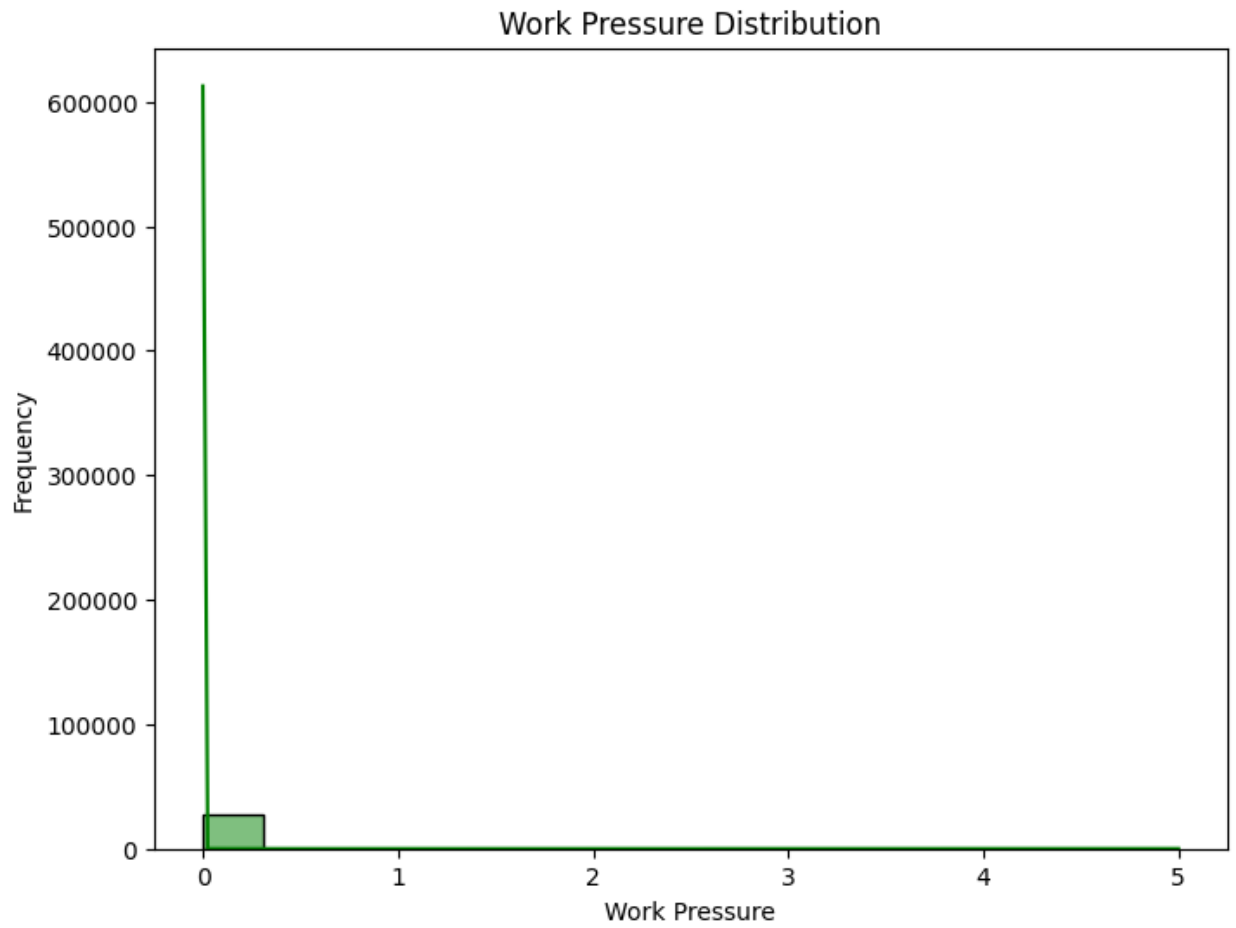
Title: Scatter Plot of CGPA and Age.

- **Box Plot:** Analyzed financial stress variation by gender to identify outliers and stress disparities [2].

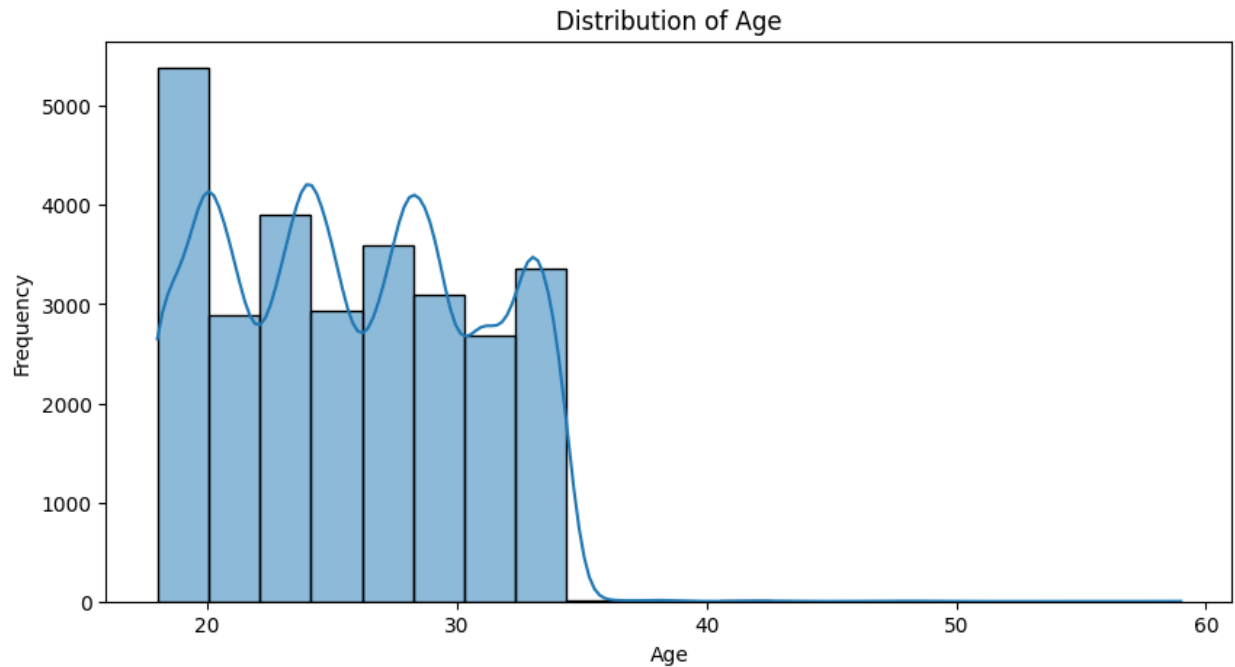


Title: Box Plot of Financial Stress by Gender.

- **Histograms:** Visualized the distribution of key features such as age and work pressure [2].

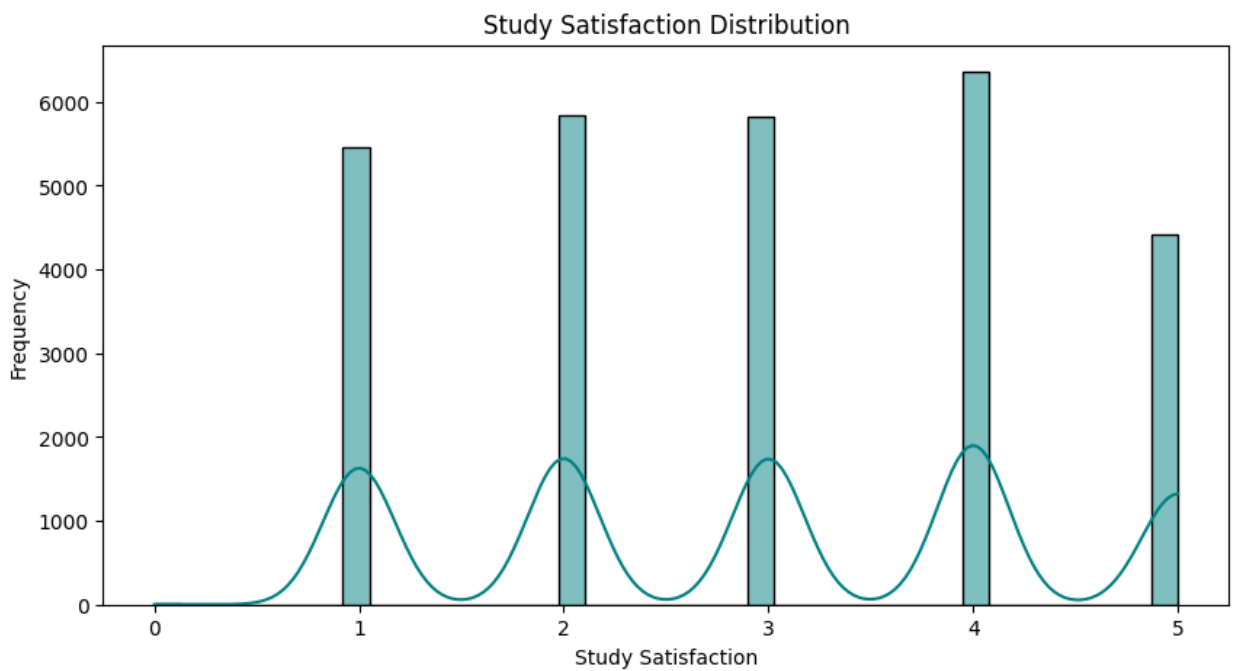


Title: Histogram Plot of Age Distribution.



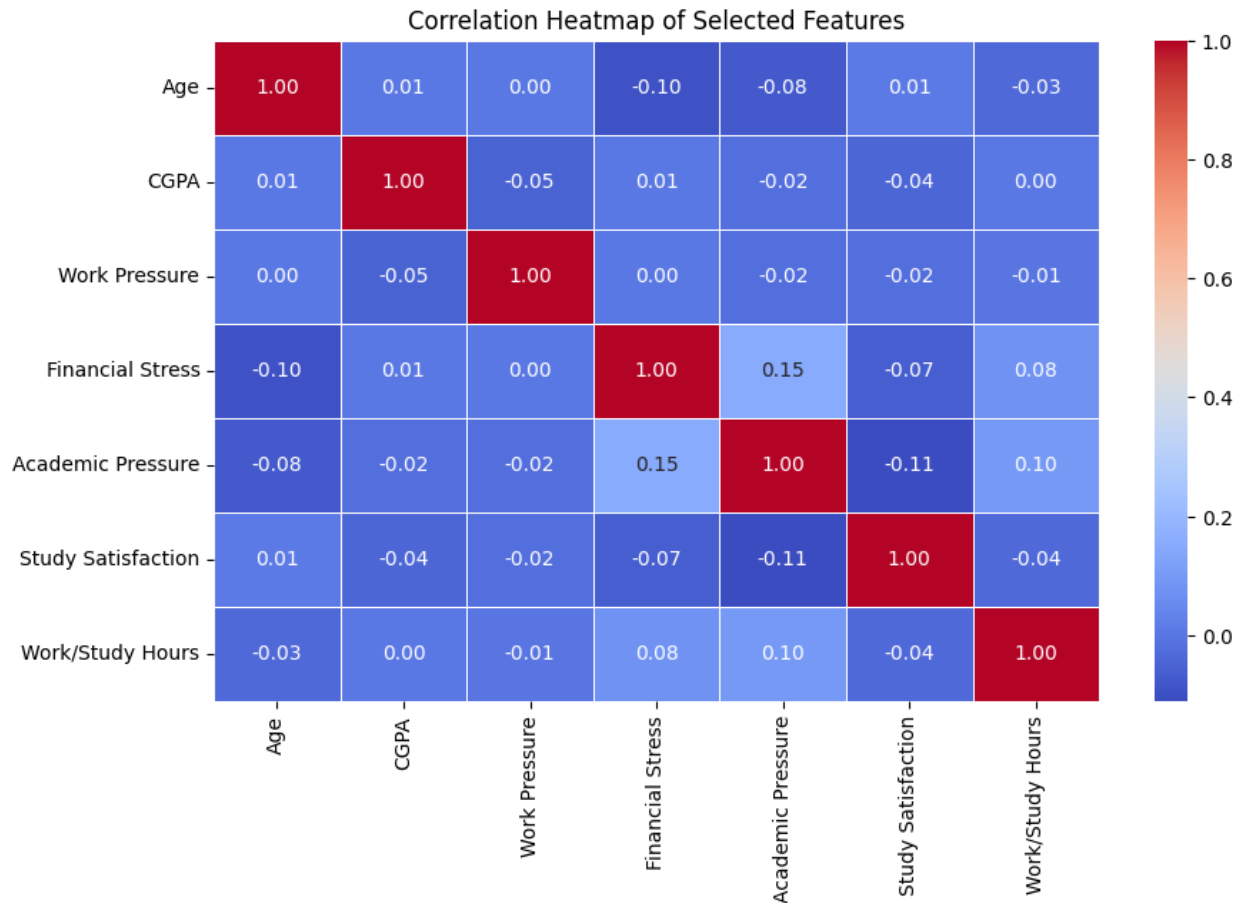
Title: Histogram Plot of Work Pressure Distribution.

- **Bar Chart:** Showed the frequency of different study satisfaction levels [2].



Title: Bar Chart of Study Satisfaction Distribution.

- **Correlation Matrix:** Identified relationships between features, highlighting financial stress and academic performance as critical predictors [2].



Title: Correlation Heatmap of Selected Features.

2.3 Model Building

A custom Logistic Regression model was implemented with the following steps:

- **Data Splitting:** Divided the dataset into training and testing sets.
- **Training:** Trained the Logistic Regression model using gradient descent.
- **Prediction:** Generated predictions on both training and test datasets.

2.4 Model Evaluation

Model performance was assessed using standard evaluation metrics:

- **Accuracy:** Proportion of correct predictions over total predictions [3].
- **Precision:** A key metric for imbalanced datasets, it measures the percentage of true positive predictions among all positive predictions [3].
- **Recall:** Indicates the model's ability to identify positive classes correctly [3].
- **F1-Score:** The harmonic mean of precision and recall provides a balanced evaluation metric, particularly when class distributions are imbalanced [3].

- **Confusion Matrix:** Provided insights into true positives, false positives, true negatives, and false negatives [3].
- **Train Accuracy:** Achieved reasonable accuracy on the training dataset.
- **Test Accuracy:** Demonstrated decent generalization on the test dataset.

3. Hyper-Parameter Optimization with Cross-Validation

Hyper-parameter optimization was conducted using cross-validation techniques to improve model performance. The following methods were implemented:

3.1 Cross-Validation

Cross-validation was employed to split the data into multiple folds for robust model evaluation [4].

3.2 Hyper-Parameter Tuning

Fine-tuned parameters such as learning rate, regularization strength, and maximum iterations to improve model performance [4].

3.3 Suggested Methods

- **GridSearchCV:** Recommended for systematic hyper-parameter tuning in future work [4].

4. Feature Selection

Manual exploration and domain knowledge were used for feature selection.

Suggested Techniques for Future Work:

- **Recursive Feature Elimination (RFE):** Systematically removes less important features [5].
- **SelectKBest:** Selects the top features based on statistical tests [5].

5. Conclusions

5.1 Key Findings

- The Logistic Regression model performed well, demonstrating decent accuracy on both training and test datasets.
- Features like financial stress and academic performance emerged as key predictors of depression levels.

5.2 Final Model

The final model based on Logistic Regression was the most effective at predicting depression levels among students.

5.3 Challenges

- **Data Quality Issues:** Missing values and inconsistent data types required significant preprocessing.

- **Feature Selection:** Identifying the most predictive features was challenging without automated selection techniques.
- **Model Evaluation Metrics:** Incorporating all standard evaluation metrics was limited.

5.4 Future Work

- **Advanced Classification Algorithms:** Implement models such as Random Forest and Gradient Boosting Classifiers.
- **Optimizing Feature Selection:** Use automated techniques for feature selection.
- **Hyper-Parameter Tuning:** Conduct GridSearchCV for optimal model performance.
- **Data Augmentation:** Expand the dataset for better generalization.
- **Incorporating Additional Features:** Explore socio-economic and lifestyle-related features impacting student mental health.

References

1. Opeyemi, S. (2023). Student Depression Dataset [Data file]. Retrieved from Kaggle.
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4. Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.
5. Kuhn, M., & Johnson, K. (2013). *Applied Predictive Modeling*. Springer.